

Relation Extraction Using BiLSTM and DeBERTa

ABHISHEK DAS¹, NAZRIN BABAYEVA², AND JUHA IM³

¹School Of Engineering, 2025 University Of Manchester, Manchester, M14 P10

**abhishek.das-2@postgrad.manchester.ac.uk*

**nazrin.babayeva@postgrad.manchester.ac.uk*

**juha.im@postgrad.manchester.ac.uk*

Compiled March 7, 2025

Relation extraction is a fundamental task in Natural Language Processing (NLP), particularly in structured information retrieval. In this paper, we investigate the performance of two Deep Learning models: ‘DeBERTa’ and ‘BiLSTM attention’, on the ‘Semeval-2010 Task 8’ dataset for relation classification. BERT, a transformer-based model, leverages contextual embeddings and self-attention mechanisms to capture deep semantic relationships, while BiLSTM, a recurrent model, processes sequential data bidirectionally to extract relational patterns. We implement and evaluate both models using standard classification metrics, analyzing their adaptations, functionality, strengths and limitations in terms of accuracy, generalization, and computational efficiency. Our experimental results contribute to the ongoing discussion on the suitability of transformer-based and recurrent architectures for relation extraction, providing insights into their comparative effectiveness in NLP applications.

<http://dx.doi.org/10.1364/ao.XX.XXXXXX>

1. INTRODUCTION

A. The task

Relation Extraction (RE) is a process in natural language processing (NLP) that involves identifying and classifying relationships between entities within a text. RE is crucial for tasks like information retrieval, knowledge graph construction, and question answering systems. It helps in structuring unstructured data, making it easier to analyze and use.

In this work, we focus on finding and classifying semantic relations in pairs of nominals in sentences. Given a sentence, annotated with nominals e1 and e2, the task is to classify nine semantic relations between these two nominals: Cause-Effect, Instrument-Agency, Product-Producer, Content-Container, Entity-Origin, Entity-Destination, Component-Whole, Member-Collection, Message-Topic, or Other if it does not belong to any of the nine annotated relations.

For example, People and downtown are connected in a Entity-Destination relation in the sentence "People have been moving back into downtown." In this instance, the relation between people and downtown could be inferred by the meaning of the two nominals and the context of "moving back" around them. Therefore, how to capture and express vocabulary and contextual information within a sentence is an important topic to be addressed for extracting semantic relationships.

We propose attention-based BiLSTM and DeBERTa to solve the relation extraction. Both are deep learning models, but attention-based BiLSTM does not use a transformer model, and DeBERTa is based on transformers. We were curious about what

difference the presence or absence of a transformer would make on the results, so we chose a model based on BERT, a transformer, and one of the LSTM types of models.

B. SemEval2010 task8

SemEval-2 Task 8 focuses on Multi-way classification of semantic relations between pairs of nominals. The task was designed to compare different approaches to semantic relation classification and to provide a standard testbed for future research.[1]

The data is drawn from a variety of domains, including news articles, Wikipedia, and other web sources. It covers a wide range of topics to ensure diversity in the examples. The dataset both text and annotations is publicly and readily available from the Hugging Face repository¹. It comes with predefined training and test subsets, which has been adopted by many researchers in NLP community, providing a standard benchmark for relation classification. However, it does not include a separate development (validation) subset. The training subset includes 8,000 examples and the test subset has 2717 examples. This size is generally sufficient to support the development of various machine learning approaches, including deep learning model. Additionally, it covers 19 types of relations, including Cause-Effect, Instrument-Agency, Product-Producer, and others.

The dataset is available in a text format, with each example consisting of a sentence and the annotated relation between a pair of nominals. Its annotations are of high quality, as the developers ensured reliability through a rigorous annotation process. As shown in Figure 1, the inter-annotator agreement (IAA) was measured to ensure the reliability of the annotations. The IAA

Relation	Freq	Pos	IAA
Cause-Effect	1331 (12.4%)	91.2%	79.0%
Component-Whole	1253 (11.7%)	84.3%	70.0%
Entity-Destination	1137 (10.6%)	80.1%	75.2%
Entity-Origin	974 (9.1%)	69.2%	58.2%
Product-Producer	948 (8.8%)	66.3%	84.8%
Member-Collection	923 (8.6%)	74.7%	68.2%
Message-Topic	895 (8.4%)	74.4%	72.4%
Content-Container	732 (6.8%)	59.3%	95.8%
Instrument-Agency	660 (6.2%)	60.8%	65.0%
Other	1864 (17.4%)	N/A ⁴	N/A ⁴
Total	10717 (100%)		

Fig. 1. Annotation Statistics. Freq: Absolute and relative frequency in the dataset; Pos: percentage of “positive” relation instances in the candidate set; IAA: inter-annotator agreement presented by Iris Wu et al.

for different relations varied, with some relations having higher agreement than others. These measures indicate that the annotations are generally reliable, with some relations being more challenging to annotate consistently.

2. RELATED WORK

SemEval2010 task8 is a standard related to relation extraction, so there has been a considerable amount of research on it. **SVM:** Rink and Harabagiu used an SVM classifier to classify relation types and directions by combining several features that capture the context, semantic role, and existing relationships of nouns. **[2] CNN:** Sentence level features are learned using a convolutional model, and concatenated with lexical features to form the final extracted feature vector. **[3] BiLSTM:** Shu et al propose bidirectional long short-term memory networks (BLSTM) to model the sentence with complete, sequential information about all words. they also use features derived from the lexical resources such as WordNet. **[4] Attention based BiLSTM:** Att-BLSTM converts input sentences into word embeddings, extracts high-level features through BiLSTM, and then generates sentence-level feature vectors through the attention mechanism to classify relationships. **[5] BERT:** A pre-trained model that efficiently encodes token spans and span pairs to simultaneously learn entities and relationships. It uses span modules to express entities and relationships while requiring fewer parameters than existing models. This model was pre-trained with a knowledge graph extracted from Wikipedia. **[6] RBERT:** Shanchuan and Yifan utilize a pre-trained BERT model to perform relation classification. By inserting special tokens before and after two target entities in a sentence, BERT captures the location information of the entities, and inputs the output embedding of BERT into a multilayer neural network to classify the relationship. **[7]**

We have found that not only supervised learning but also many neural network models and transformer models have been applied to solve relation extraction. Our task is also to find potential relations between given nouns and learn effective features at the lexical and sentence levels. Therefore, we will mine sentence-level features using BiLSTM and DeBERTa and extract relations from sentences to find out the effect of the presence or absence of transformers.

3. METHODOLOGY

A. Attention based BiLSTM

This model builds upon a BiLSTM (Bidirectional Long Short-Term Memory) with an attention mechanism to classify relationships between entities in a sentence. Unlike traditional BiLSTMs, which process the sequence in both forward and backward directions with equal knowledge of the entire sentence, this model introduces an asymmetrical structure. The first LSTM processes the sentence from left to right, completely unaware of what comes next, while the second LSTM starts from the end of the sentence, but with prior knowledge of the first half. (Fig.??) This unique setup allows the backward LSTM to leverage additional contextual information, improving the model’s ability to capture long-range dependencies and disambiguate entity relationships.

To ensure the dataset is properly structured, a preprocessing pipeline was implemented. The dataset consists of sentences with marked entities and their respective relationship labels. Entity markers such as <e1> and <e2> tags were preserved to maintain clarity, while redundant spaces were removed, and entity positions were adjusted for consistency. Sentences were tokenized, and additional markers [E1] and [E2] were introduced around entities to further reinforce their positions. Relation labels were mapped to numerical IDs, and inputs were padded or truncated to a fixed length of 128 tokens. These processed data instances were stored as [8] Torch tensors, ensuring compatibility with deep learning models. The dataset was then split into 80

To enhance the model’s semantic understanding, pre-trained [9] GloVe embeddings (300-dimensional) were integrated. These embeddings capture word meanings based on their co-occurrence patterns in large text corpora, providing the model with a richer linguistic foundation. A specialized GloVe embedding loader mapped words from the dataset to their corresponding pre-trained vectors, while words not found in the GloVe vocabulary were initialized with small random values in the range [-0.05, 0.05]. The resulting embedding matrix was stored as a PyTorch tensor and used to initialize the model’s embedding layer.

At the heart of the model is the BiLSTM network, which operates with an asymmetrical processing structure mentioned before in this section: the forward LSTM reads the sentence from beginning with zero knowledge while the backward LSTM starts from the last token but has access to the outputs of the forward LSTM. This design ensures that the latter half of the sentence, which often provides crucial context, can refine the understanding of the first half.

To further improve the model’s interpretability and effectiveness, an attention mechanism was applied. A linear transformation assigns attention scores to each LSTM time step, which are then normalized using a [10] softmax Function to determine attention weights. These weights highlight the most relevant words in the sentence, and a weighted sum of the LSTM outputs is computed, allowing the model to focus on the most informative parts of the sequence.

For training, the model was optimized using AdamW with a learning rate of 1e-3, and cross-entropy loss was used to handle multi-class classification. The training process spanned 10 epochs, leveraging [11] GPU acceleration when available. The best-performing model was selected based on validation accuracy, ensuring optimal performance. For evaluation, accuracy and [12] Weighted F1-score were used as key performance metrics, while qualitative analysis was conducted by manually inspecting test samples to assess real-world applicability. By

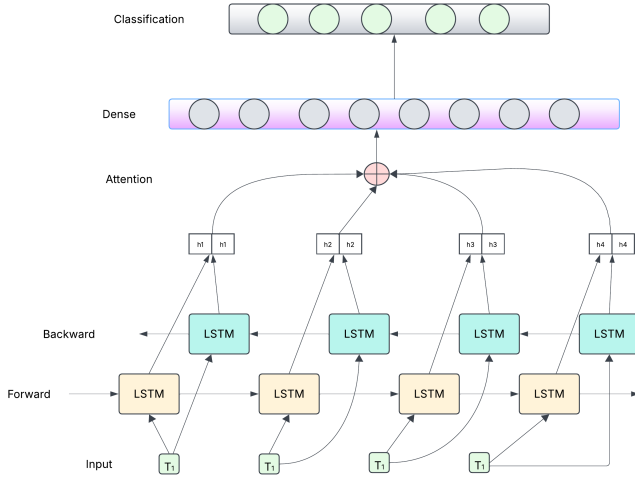


Fig. 2. Architecture of Attention Based BiLSTM

integrating an innovative asymmetrical BiLSTM structure, attention mechanisms, and GloVe embeddings, this model effectively captures entity relationships with enhanced contextual awareness.

B. DeBERTa

DeBERTa (Decoding-enhanced BERT with Disentangled Attention) was introduced by Pengcheng He et al. in 2020 [13] as an improvement over BERT and RoBERTa. It enhances contextual representation learning through a **disentangled attention mechanism** and an **enhanced masked decoder**, enabling more effective dependency modeling across token embeddings. These improvements address limitations in traditional self-attention by separately encoding content and position information, allowing for a richer contextual understanding.

In this work, we customize the DeBERTa-base model for relation extraction (RE) by incorporating an entity-aware representation strategy inspired by the **RBERT** (Relation-aware BERT) framework proposed by Shanchan Wu et al. in 2019 [6]. Standard relation extraction models typically rely on the final hidden state of the [CLS] token for classification, under the assumption that it encapsulates the overall sentence context. However, Wu et al. demonstrated that integrating entity representations alongside the [CLS] token yields richer contextual understanding and improves performance in relation classification tasks.

Following this approach, we extract the contextualized representations of both entity mentions from the final hidden states corresponding to their token spans. Instead of treating each entity independently, we aggregate the attention output from multiple layers across all subword tokens belonging to an entity span. Specifically, as shown in Figure 3, the output representations from heads H_i to H_j are averaged to obtain a compact representation for the first entity. Similarly, the output from heads H_k to H_m is averaged for the second entity. These aggregated entity representations are then concatenated with the [CLS] token representation to form a comprehensive feature vector. The concatenated representation is passed through a feedforward neural network followed by a softmax classifier to predict the relation label. This modification theoretically enhances the model's ability to capture fine-grained semantic interactions, leading to improved accuracy and F1 scores compared to the standard DeBERTa model. Our input sentences are sourced

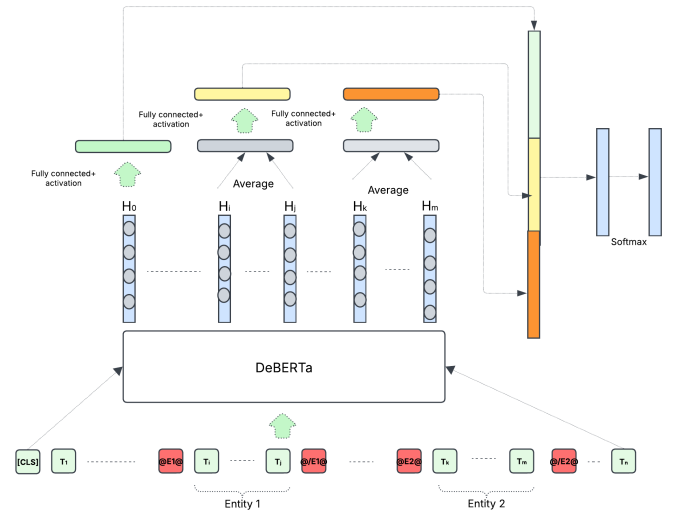


Fig. 3. Model Architecture inspired from the architecture presented by Shanchan Wu et al.

from the **SemEval-2010 Task 8** dataset, which is widely used for benchmarking relation extraction models. Each sentence is preprocessed into a structured dictionary containing the sentence text, entity start and end positions, and the corresponding relation label. To explicitly mark entity spans, we enclose the first entity within the **@E1@**, **@/E1@** tags and the second entity within the **@E2@**, **@/E2@** tags before tokenization. The sentences are then tokenized using the DeBERTa tokenizer, ensuring that entity markers are preserved in the input token sequence.

For efficient training, we construct a **TensorDataset** containing the tokenized inputs, attention masks, entity position indices, and relation labels. This dataset is split into training and validation sets and subsequently loaded into **DataLoader** objects for batch processing. The hyperparameters used during training are detailed in Table 1. Through empirical evaluation, we observed that the model begins overfitting after approximately four to five epochs, as reflected in the stagnation of the validation F1 score despite improvements in training performance. To mitigate overfitting, we introduced a dropout rate of **0.1**, which helped stabilize the model and led to an approximate **0.2% increase in validation accuracy**. Additionally, extensive hyperparameter tuning revealed that the optimal learning rate for our transformer-based model was consistently **2e-5**, a finding that aligns with prior studies on fine-tuning large pre-trained models. This may be attributed to the intrinsic learning dynamics of transformer architectures and the dataset's relatively small size.

For comparison, we also evaluated several other pre-trained transformer-based models, including **BERT**, **RoBERTa**, **ALBERT**, and **DistilBERT**. Our experimental results indicate that these models achieve performance levels similar to our RDeBERTa implementation. This suggests that while architectural modifications such as disentangled attention improve contextual encoding, relation extraction tasks remain highly dependent on dataset characteristics and fine-tuning strategies.

4. EVALUATION AND RESULTS

We decided to use F1-score as a metric to evaluate the trained relation extraction model. F1-score is a harmonic mean of precision

Learning Rate	$2e^{-5}$
Dropout	0.1
Epochs	7
Batch Size	32

Table 1. Model Hyperparameters

and recall, which is useful for balancing the two metrics. It simultaneously considers the proportion of models that accurately predict (precision) and how well they find the actual relations (recall). In addition, F1-score provides better evaluation for imbalanced data. In relation extraction tasks, certain relations may appear more frequently than others. Finally, SemEval-2010 Task 8 uses macro-average F1-score as the official evaluation metric.

A. Attention based BiLSTM

Initially, we implemented a basic Bidirectional LSTM (BiLSTM) model. However, the F1 score was unsatisfactory at 0.4. To address this, we incorporated an attention mechanism, which improved the F1 score to 0.59. Further enhancement was achieved by integrating GloVe 300-dimensional pretrained embeddings, resulting in an F1 score of 0.67. The detailed results are presented in Table 2.

Model	Feature	F1
BiLSTM	Implemented a vanilla bidirectional LSTM using keras library.	0.4
Attention Based BiLSTM	Added Attention in order to better capture the token dependencies.	0.59
Attention based BiLSTM with pre-trained word embeddings	Used glove 300 dimensional pretrained embeddings as the embedding matrix.	0.67

Table 2. Performance change of BiLSTM

B. DeBERTa

As shown in Table 3, We initially experimented with BERT, RoBERTa, and ALBERT for relation extraction; however, the performance did not meet our expectations. In contrast, DeBERTa demonstrated significantly better performance, prompting us to focus on developing this model further.

Our first enhancement involved integrating BiGRU, which resulted in an F1 score of 0.78. Subsequently, we experimented with additional attention mechanisms, but this approach reduced the F1 score to 0.77. Seeking alternative methods, we reviewed literature related to R-BERT and applied its principles, achieving an F1 score of 0.80, which aligned with our performance expectations.

5. DISCUSSION AND CONCLUSION

As shown in the Table 4, we can see that the F1 score of Atten-BiLSTM is relatively low. However, we confirmed that the training time for the model was three times longer for DeBERTa than

Model	Feature	F1
DeBERTa with BiGRU	DeBERTa extracts deep semantic features, which are then processed by a BiGRU to capture forward and backward dependencies.	0.78
DeBERTa with BiGRU and attention	Added an attention layer in order to better capture the dependencies.	0.77
RDeBERTa	Concatenated the [CLS] token with the two entity representations.	0.80

Table 3. Performance change of DeBERTa

Model	F1
Atten-BiLSTM	0.63
DeBERTa	0.80

Table 4. Final results

for Atten-BiLSTM (training environment: Google Colab GPU). In addition, DeBERTa's model was more complex and required more libraries to be imported, and errors often occurred during the training process.

A. Attention based BiLSTM

Strengths: Atten-BiLSTM captures the bidirectional contextual information of a sentence by processing the input sequence both forward and backward. This allows us to better understand the overall meaning of the sentence. In addition, the LSTM structure can handle long-term dependencies well even in long sequences, so it can maintain important information in the sentence without losing it. Adding an attention mechanism allows the model to focus more on important words, which improves performance in tasks such as relation extraction.

Weakness: combining BiLSTM with attention mechanisms can increase computational cost, which can slow down training and inference speed. Hyperparameter tuning can be difficult as the model architecture becomes more complex. Memory usage increases when processing long sequences, which can make it difficult to process large datasets.

B. DeBERTa

Strengths: DeBERTa uses a disentangled attention mechanism to separate the content and position information of words to generate more sophisticated contextual representations. DeBERTa shows excellent performance on various natural language processing tasks. In addition, as a model pre-trained on a large-scale dataset, it can be easily applied to various downstream tasks.

Weakness: However, DeBERTa is a large-scale model, requiring a lot of computational resources and memory for training and inference. In addition, the model structure is complex, making hyperparameter tuning and model optimization difficult. Finally, due to the high computational cost and memory usage, it may be difficult to use in real-time applications.

REFERENCES

1. I. Hendrickx, S. N. Kim, Z. Kozareva, *et al.*, "SemEval-2010 task 8: Multi-way classification of semantic relations between pairs of nominals," in *Proceedings of the 5th International Workshop on Semantic Evaluation*, K. Erk and C. Strapparava, eds. (Association for Computational Linguistics, Uppsala, Sweden, 2010), pp. 33–38.
2. B. Rink and S. Harabagiu, "UTD: Classifying semantic relations by combining lexical and semantic resources," in *Proceedings of the 5th International Workshop on Semantic Evaluation*, K. Erk and C. Strapparava, eds. (Association for Computational Linguistics, Uppsala, Sweden, 2010), pp. 256–259.
3. D. Zeng, K. Liu, S. Lai, *et al.*, "Relation classification via convolutional deep neural network," in *Proceedings of COLING 2014, the 25th International Conference on Computational Linguistics: Technical Papers*, J. Tsujii and J. Hajic, eds. (Dublin City University and Association for Computational Linguistics, Dublin, Ireland, 2014), pp. 2335–2344.
4. S. Zhang, D. Zheng, X. Hu, and M. Yang, "Bidirectional long short-term memory networks for relation classification," in *Proceedings of the 29th Pacific Asia Conference on Language, Information and Computation*, H. Zhao, ed. (Shanghai, China, 2015), pp. 73–78.
5. P. Zhou, W. Shi, J. Tian, *et al.*, "Attention-based bidirectional long short-term memory networks for relation classification," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, K. Erk and N. A. Smith, eds. (Association for Computational Linguistics, Berlin, Germany, 2016), pp. 207–212.
6. J. Li, Y. Katsis, T. Baldwin, *et al.*, "Spot: Knowledge-enhanced language representations for information extraction," in *Proceedings of the 31st ACM International Conference on Information & Knowledge Management*, (Association for Computing Machinery, New York, NY, USA, 2022), CIKM '22, p. 1124–1134.
7. S. Wu and Y. He, "Enriching pre-trained language model with entity information for relation classification," in *Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, (Association for Computing Machinery, New York, NY, USA, 2019), CIKM '19, p. 2361–2364.
8. A. Paszke, S. Gross, F. Massa, *et al.*, *PyTorch: an imperative style, high-performance deep learning library* (Curran Associates Inc., Red Hook, NY, USA, 2019).
9. J. Pennington, R. Socher, and C. Manning, "GloVe: Global vectors for word representation," in *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, A. Moschitti, B. Pang, and W. Daelemans, eds. (Association for Computational Linguistics, Doha, Qatar, 2014), pp. 1532–1543.
10. S. Totaro, A. Hussain, and S. Scardapane, "A non-parametric softmax for improving neural attention in time-series forecasting," *Neurocomput.* **381**, 177–185 (2020).
11. E. A. Träff, A. Rydahl, S. Karlsson, *et al.*, "Simple and efficient gpu accelerated topology optimisation: Codes and applications," *Comput. Methods Appl. Mech. Eng.* (2023).
12. K. Leung, "Micro, macro weighted averages of f1 score, clearly explained," (2023). Accessed: 2025-03-06.
13. P. He, X. Liu, J. Gao, and W. Chen, "Deberta: Decoding-enhanced bert with disentangled attention," *ArXiv abs/2006.03654* (2020).